Week 2: Representation – Bayes Networks
Practical Assignment

In this practical, we look at three simple problems: the Monty-Hall problem, the student network, and the casino hidden Markov network (HMM). For each problem, we use a BN to model the problem, we construct the model, and we then query the model. We also use each model to investigate the flow of influence between nodes (i.e., we investigate (conditional) dependence and independence between RVs in the model).

1 The Monty-Hall problem

For this one we are going to consider the well-known Monty-Hall problem. Let's first state the original problem: In a TV game show, a participant must choose between one of three doors. Behind one of the doors there is an expensive car; the other two conceal somewhat smelly goats. The participant indicates a certain door, after which the game-show host (Monty Hall) reveals a goat behind another door. You can assume that the host will never at this stage open the initial choice of the participant or the door with the car. Except for this, he has no preference for any specific door. The participant may now reconsider his choice. Under the assumption that he actually wants to win the car, should the player stick with his original choice or should he change to the other remaining door?

Model the above situation with a BN. Use three random variables:

1. I is the initial choice of the participant – it takes on values $0, 1$ or 2 depending on his door of choice.
2. C is where the car is – once again it takes on values $0, 1$ or 2 depending on where the car actually is.
3. M is the door that Monty opens – once again it takes on values $0, 1$ or 2.

If we reason causally, then it should be clear that the initial choice I and the location of the car C are not influenced by any other RVs, and the door that Monty opens, M, is influenced by both I and C. An appropriate Bayes net structure is therefore $I \rightarrow M \leftarrow C$. Another way of getting to this same structure is to use the general factorisation

$$p(M, I, C) = p(M|I, C)p(C, I),$$

and then realise that the location of the car C is independent of the initial choice I, so we can say that $p(C, I) = p(C)p(I)$. The Bayes net structure $I \rightarrow M \leftarrow C$ and the factorisation $p(M, I, C) = p(M|I, C)p(C)p(I)$ encode the same independence assumption. To define this network, we now have to set up the distributions $p(I)$, $p(C)$, and $p(M|I, C)$. The last one, $p(M|I, C)$, is the tricky one to set up – carefully include all the considerations that Monty have to keep in mind when doing so.

1.1 Solve the Monty-Hall problem

With this in place you can now easily answer the basic Monty-Hall question with `emdw` using the factor definitions and factor operations introduced in the previous practical.

(a) Construct the factors in `emdw` and then calculate the joint distribution $p(M, I, C)$.

(b) Pick an initial choice (e.g. $I = 0$) and a (valid) door that Monty opened (e.g. $M = 1$) and use factor operations to get the belief distribution over the location of the car given the evidence, or $p(C|, I = i, M = m)$. That should tell you that it is better for the player to switch to the other door. Does this make sense to you?

With that out of the way, lets more specifically focus on the concept of flow of influence.

1.2 Flow of influence via collider nodes

Use your code of Question 1.1 and the factor operations introduced in the previous practical to investigate the following using `emdw`.

(a) The path $I \to M \leftarrow C$ forms a collider node at M (also called a *v-structure*). We want to determine whether I and C are statistically dependent or independent given that M is either unobserved or observed (i.e., we want to know under which conditions "influence flows between I and C"). First test your intuition: If M is unobserved, does knowledge of I tell you anything about C? And what if M is observed?

(b) Now let's investigate it using `emdw`:

- Set I to one of its allowed values, while leaving the other variables unobserved. Does this change the marginal belief over C? What is your conclusion about the dependence (or not) of C on I when M is unobserved?[1] Is this consistent with d-separation (i.e., are I and C d-separated (given no other RVs)?).
- Set M to one of the values it can take on, while leaving the other variables unobserved. Find the marginal belief over C. Now, with M still on the value you chose, set I to one of the two remaining values it can legally take on. Does this change the marginal belief over C?[2] Is this consistent with d-separation (i.e., are I and C d-separated by M?)

(c) Let's make things slightly more complex. The game remains more or less as it was, but we are now being prevented from observing M directly. Instead, we get to observe a variable R that (unreliably) reports which door Monty opened[3]. This will introduce a node R and a single edge $M \to R$ to the Bayes net (since R is influenced by M, but conditionally independent of I and C given M – can you see it?) Define the factor $p(R|M)$ such that we have a 80% chance that R will report the correct value of M, and 10% chances that it will (erroneously) report one of the other two values for M. Implement the new factor and construct the new joint distribution $p(R, M, I, C)$.

(d) We now want to investigate whether influence flows between I and C with R observed/unobserved (i.e., M is never directly observed, but only indirectly observed via R). Test your intuition again: If R is unobserved, does knowledge of I tell you anything about C? And what if R is observed?

(e) Repeat (b), but now with R observed/unobserved instead of M. What are your conclusions, and do they correspond to the theory of d-separation?

(f) Can you specify other Bayes nets that encode the same conditional independence assumptions by reversing the direction of some of the arrows (i.e., is there another BN that is Markov equivalent to this one)? Motivate your answer.

(g) What happens to C when you set the observed value of R to be the same as I? Can you make sense of this?

2 The student network

The student network is an example problem that Daphne Koller uses to illustrate BNs. This network is also described in the **PGM Tutorial** Figure 3.1 and **Murphy** Section 4.2.2.2. If you have not done so already, study this example in the references.

2.1 Construct and query the model

(a) Construct the factors of the model in `emdw` and calculate the joint distribution $p(D, I, G, L, S)$.

(b) Using factor operations, calculate the belief that the student will receive a good recommendation letter given that they got a good SAT score and that the course was difficult, or $p(L|D = 1, S = 1)$[4]. Does the answer make sense to you in the context of the problem?

[1] If $p(C|I = 0) = p(C)$, does this mean that $C \perp\!\!\!\perp I$, or do you have to test *all* values of I? But if $p(C|I = 0) \neq p(C)$, can you say that C is dependent of I, or do you also have to test all values of I?

[2] Similarly to the previous point, do you have to test *all* values of M to make a reliable conclusion about the conditional (in)dependence of C and I given M?

[3] I.e., this is *unreliable* evidence as discussed in Barber Section 3.2.2.

[4] The idea of this question is to use `emdw`. However, I recommend that you repeat this question by hand at home – it is a good opportunity to practice these numerical manipulations and would also clarify the concepts involved.

2.2 Flow of influence via confounder nodes

In the structure $G \leftarrow I \rightarrow S$, I is a parent of G and S. This is a so-called *fork* structure, and I is called a *confounder*. We want to investigate whether influence flows between G and S with I either observed or unobserved.

(a) Test your intuition: Does knowledge of a student's SAT score S tell you anything about the student's grade G when the student's intelligence I is unobserved? And what if the intelligence I is observed?

(b) Similarly to Question 1.2(b), use `emdw` to investigate the (conditional) independence between G and S when I is observed/unobserved. What are your conclusions, and do they correspond to the theory of d-separation?

3 The casino HMM

The casino HMM is an example in **Murphy** Section 9.2.1.1 that describes an occasionally dishonest casino. If you have not done so already, study this example in **Murphy**.

3.1 Construct and query the model

(a) The factors required by the model are $p(z_t|z_{t-1})$, $p(y_t|z_t)$, and $p(z_1)$ (for $t \in [1, T]$). Write these three factors on paper as discrete tables. Note that the problem description in **Murphy** does not give us enough information to specify $p(z_1)$; you can assume that the probability that the casino starts (at $t = 1$) with a fair die is 0.5.

(b) For this question, we only look at 3 time steps, which means that we use the BN shown in **Murphy** Figure 9.1. Construct the joint distribution $p(z_1, z_2, z_3, y_1, y_2, y_3)$ using factor operations in `emdw`.

(c) Let's think about storage complexity for a moment: If no default value is used for the discrete table (i.e., the table is not sparsified), how many entries does the table representing the joint distribution in (b) have? If the model covers T time steps, how many entries would the table have? Can you see that it would quickly become impossible to construct and store the joint distribution as the number of time steps increase?[5]

(d) Suppose the sequence of observations is $(y_1, y_2, y_3) = (6, 5, 6)$. Now use `emdw` to calculate the posterior beliefs $p(z_1|y_1 = 6, y_2 = 5, y_3 = 6)$, $p(z_2|y_1 = 6, y_2 = 5, y_3 = 6)$ and $p(z_3|y_1 = 6, y_2 = 5, y_3 = 6)$[6]. Do these beliefs make sense to you?

3.2 Flow of influence via chain nodes

The structure $z_1 \rightarrow z_2 \rightarrow z_3$ is called a *chain* or *cascade*, and z_2 is called a *mediator*. We want to investigate whether influence flows between z_1 and z_3 with z_2 either observed or unobserved[7].

(a) Test your intuition: Does knowledge of z_3 tell you anything about z_1 when z_2 is unobserved?[8] And what if z_2 is observed?

(b) Similarly to Question 1.2(b), use `emdw` to investigate the (conditional) independence between z_3 and z_1 when z_2 is observed/unobserved. What are your conclusions, and do they correspond to the theory of d-separation?

[5] In later weeks, we will see how we can bypass the need to construct the full joint distribution, which will alleviate this issue.

[6] Which HMM algorithm that you've encountered before answers the same query as in this question?

[7] That is, we are investigating the *Markov assumption* of this first-order HMM.

[8] Note that it is easier to reason "in the causal direction"; i.e., whether knowledge of z_1 tell you anything about z_3 when z_2 is unobserved. Make sure you can reason "against the causal direction".

4 Evaluation: Write and submit a short report

Document the following in a short report:

- Question 1.1 (a) and (b) [2 marks]
- Question 1.2 (e) [2 marks]
- Question 2.1 (b) [1 mark]
- Question 2.2 (b) [2 marks]
- Question 3.1 (d) [1 mark]
- Question 3.2 (b) [2 marks]

Submit both the report (in PDF format) and all your code for this assignment on STEMLearn by **Sunday 22 February at 23:59**.

See the DE424 info sheet for general guidelines for the practical report. Also so the document about how to avoid academic misconduct in DE424 on STEMLearn. Specific instructions:

- The text in the report should be typed, the page limit is **5 pages**, and the report must be submitted in PDF format.
- At the top of the report, you should include a declaration that the report and the work presented in it are your own work. If you received any help or information from any source that contributed to the work or report, you should also mention the source and the extent of the help or information in the declaration.
- The report should be clear, but you do not have to spend much time on making it beautiful. You can, for instance, draw diagrams or write mathematical manipulations by hand, scan it in, and include it in your report as an image (instead of drawing diagrams in applications such as Inkscape or typing the mathematical manipulations as equations in LaTeX).
- You should give enough context in your report so that someone who is familiar with the module can get a good idea of what you did (and why) without having to look at your code. Conversely, you should be concise: avoid giving lengthy discussions of obvious matters, showing minute steps in mathematical manipulations, or copying large parts of your code into your report.